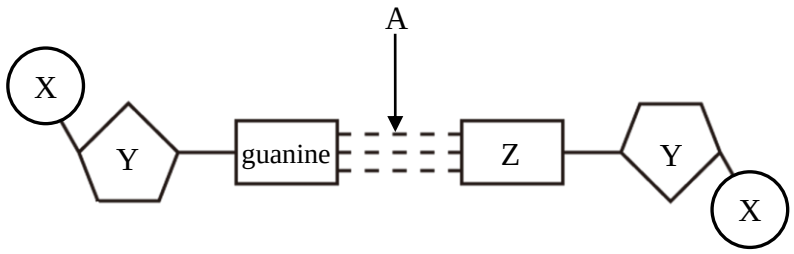
Short questions

|!|BCESC0020000001|!|

The diagram below shows the molecular structure of a part of a DNA molecule.



(a) Identify the components X, Y and Z. (3 marks)

(b) Name the basic units which make up a DNA molecule. (1 mark)

(c) What are the bonds labelled A? How is this type of bond important to DNA molecules? (2 marks)

##

(a) X is a phosphate group. (1)
Y is a five-carbon sugar / deoxyribose. (1)
Z is the base cytosine. (1)

(b) nucleotide (1)

(c) hydrogen bonds (1)
The hydrogen bonds formed between the base pairs help stabilize the double helix structure of DNA. (1)

(6 marks)

##

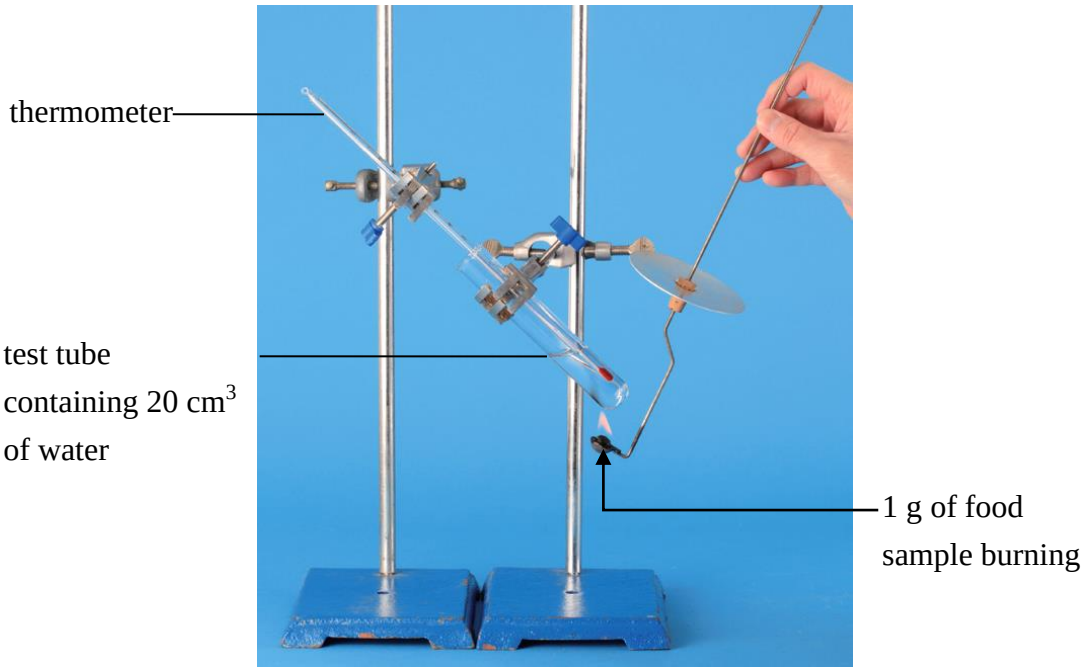
	Skills assessed	Marks
(a)	Analysing data	3
(b)	Understanding basic concepts	1

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(c)	Understanding basic concepts	2

|!|BCESB0020000002|!| ★ SI

The set-up below was used to find the energy content of three food samples (A, B and C). One gram of food sample was burned and the energy released heated the water in the test tube. The starting temperatures and the highest temperatures of the water were measured.



(a) State ***two*** variables that have to be kept constant during the investigation. (2 marks)

_____

_____

The table below shows the results.

Food sample	Water temperature (°C)		Rise in temperature (°C)	Energy content (kJ per g)
	Starting	Highest		
A	25	34	9	7.56
B	24	40	16	13.44
C	22	33		

The energy content of the food (kJ per g) can be calculated using the formula:

$$\text{Energy content} = \text{volume of water (cm}^3\text{)} \times \text{temperature rise} \times 0.042$$

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(b) Complete the table to show the rise in temperature and energy content of sample C. (2 marks)

(c) The three samples are peanut oil, rice and egg white. It is known that sample A is egg white. Based on the results, deduce which sample is peanut oil. (2 marks)

_____

_____

(d) Which element is always found in proteins but not in lipids or carbohydrates? (1 mark)

_____

##

(a) volume of water in the test tube / mass of food used / distance between the burning food and the test tube / other reasonable answer (Any ***two***, 2)

(b) Rise in temperature: 11 ^{o}C (1)

Energy content: 9.24 kJ per g (1)

(c) Sample B is peanut oil. (1)

The energy content of lipids is higher than / is roughly twice that of proteins. (1)

(d) nitrogen (1)

(7 marks)

##

	Skills assessed	Marks
(a)	Practical skills	2
(b)	Mathematical skills	2
(c)	Analysing data	2
(d)	Understanding basic concepts	1

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|!|BCESB0020000003|!| ★

(a) The table below lists statements about four types of carbohydrates. In each box, put a '✓' if the statement is true, or a '×' if it is not true. (3 marks)

	Glucose	**Starch**	**Glycogen**	**Cellulose**
Contains bonds formed by condensation				
Soluble in water				
Found in plant cells				

(b) Describe how you would test for the presence of starch in a given sample.(2 marks)

_____

_____

##

(a) 1 mark for each correct row (3)

	Glucose	**Starch**	**Glycogen**	**Cellulose**
Contains bonds formed by condensation	×	✓	✓	✓
Soluble in water	✓	×	×	×
Found in plant cells	✓	✓	×	✓

(b) Add iodine solution to the sample. (1)

It turns blue-black in the presence of starch. (1)

(5 marks)

##

	Skills assessed	**Marks**
(a)	Understanding basic concepts	3
(b)	Practical skills	2

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|!|BCESB0020000004|!| ★ SI

Sucrose is a disaccharide made up of two hexose sugars. A sucrose molecule can be broken down by the enzyme, sucrase, to give a molecule of glucose and a molecule of fructose.

$$\text{sucrose} + \text{water} \xrightarrow{\text{sucrase}} \text{glucose} + \text{fructose}$$

(a) Name the reaction that breaks down sucrose into glucose and fructose. (1 mark)

(b) What is the molecular formula of sucrose? (Hint: Hexose is a monosaccharide with six carbon atoms. The chemical formula of hexose is $C_6H_{12}O_6$.) (1 mark)

(c) In an investigation, a solution containing the enzyme sucrase was added to a sucrose solution in a test tube. The mixture was incubated at 37 °C for one hour.

(i) A sample was removed from the tube at the beginning of the investigation. It gave positive result with protein test. Explain why. (1 mark)

(ii) Another sample was removed from the tube after one hour. Describe how you would use a biochemical test to show that this sample contains reducing sugars. (2 marks)

##

(a) hydrolysis (1)

(b) $C_{12}H_{22}O_{11}$ (1)

(c) (i) The positive result is due to the presence of sucrase, which is protein in nature. (1)

(ii) Heat sample B with Benedict's solution. (1)
The formation of a brick-red precipitate would show that the sample contained reducing sugars. (1)

(5 marks)

##

	Skills assessed	Marks
(a)	Analysing data	1

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(b)	Applying concepts	1
(c)(i)	Applying concepts	1
(c)(ii)	Practical skills	2

|!|BCESC0020000005|!| **SI**

A food sample was tested for the presence of different biomolecules. The following table shows the results.

Test	Result
Benedict's test	blue solution
Iodine test	blue-black solution
DCPIP test	blue solution

(a) According to the results, state the biomolecule(s) that is/are present in the food sample. (1 mark)

_____

(b) The food sample actually contains a type of disaccharide. What is this disaccharide? (1 mark)

_____

(c) Enzyme X is an enzyme that can catalyse the breakdown of the disaccharide stated in (b).

(i) What kind of biomolecule is enzyme X? (1 mark)

_____

(ii) How could you test for the presence of enzyme X? (2 marks)

_____

_____

(iii) After the breakdown of the disaccharide catalyzed by enzyme X, what would be the product(s) formed? (1 mark)

_____

##

(a) starch (1)

(b) sucrose (1)

(c) (i) protein (1)

(ii) Dip the test end of a protein test strip into a sample. (1)

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A distinctive colour change in the test end indicates the presence of proteins. (1)

(iii) glucose and fructose (1)

(6 marks)

##

	Skills assessed	Marks
(a)	Analysing data	1
(b)	Applying concepts	1
(c)(i)	Understanding basic concepts	1
(c)(ii)	Practical skills	2
(c)(iii)	Understanding basic concepts	1

|!|BCESC0020000006|!|

Fill in the blanks to complete the following paragraph. (4 marks)

Water is essential to organisms. As it is a good (a)_____, it helps in carrying substances to different parts of the body. Besides, water is a (b)_____ in photosynthesis for synthesizing carbohydrates.

Water absorbs a lot of heat with only a slight change in its temperature. Because of this property, oceans have relatively (c)_____ temperatures and thus provide a suitable habitat for aquatic organisms. Moreover, water has a high (d)_____ and so it provides buoyancy for aquatic organisms.

##

(a) solvent (1)

(b) reactant (1)

(c) stable / constant (1)

(d) density (1)

(4 marks)

##

	Skills assessed	Marks
(a)	Understanding basic concepts	1
(b)	Understanding basic concepts	1
(c)	Understanding basic concepts	1
(d)	Understanding basic concepts	1

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|!|BCESB0020000007|!| ★

For each of the function of water listed in column 1, select from column 2 one example that matches it. Put the appropriate letter in the space provided. (3 marks)

Column 1			***Column 2***	
as a transport medium	(a)	_____	A.	sweat
as a supporting agent	(b)	_____	B.	phloem sap
as a cooling agent	(c)	_____	C.	urine
			D.	aqueous humour

##

(a) B (1)

(b) D (1)

(c) A (1)

_____

(3 marks)

##

	Skills assessed	Marks
(a)	Understanding basic concepts	1
(b)	Understanding basic concepts	1
(c)	Understanding basic concepts	1

|!|BCESC0020000008|!|

Phosphorus is one of the elements required by living organisms. It can be used to form different chemical substances.

(a) (i) Name ***two*** organic substances in organisms that are composed of phosphorus. (2 marks)

_____

_____

(ii) State ***one*** function of the two organic substances in organisms. (2 marks)

_____

_____

(b) (i) Name ***one*** inorganic substance in organisms that is composed of phosphorus. (1 mark)

_____

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(ii) State ***one*** function of the inorganic substances in organisms. (1 mark)

##

(a) (i) phospholipids / nucleotides / ATP / other reasonable answer (Any ***two***, 2)

(ii) Any ***two*** of the corresponding functions: (2)

- Phospholipids are the main components of cell membranes.
- Nucleotides form nucleic acids which govern the inheritance of characteristics in organisms.
- ATP is an energy carrier / helps energy transfer in cells.

(b) (i) calcium phosphate / other reasonable answer (1)

(ii) Calcium phosphate is a constituent of bones and teeth in mammals. (1)

(6 marks)

##

	Skills assessed	Marks
(a)(i)	Understanding basic concepts	2
(a)(ii)	Understanding basic concepts	2
(b)(i)	Understanding basic concepts	1
(b)(ii)	Understanding basic concepts	1

|!|BCESC0020000009|!|

For each of the carbohydrates listed in column 1, select from column 2 one phrase that matches it. Put the appropriate letter in the space provided. (3 marks)

Column 1			***Column 2***	
sucrose	(a)	_____	A.	a storage form of carbohydrate in mammals
cellulose	(b)	_____	B.	a transporting form of carbohydrate in plants
glycogen	(c)	_____	C.	a transporting form of carbohydrate in mammals
			D.	a component of cell wall in plant cells

##

(a) B (1)

(b) D (1)

(c) A (1)

(3 marks)

##

	Skills assessed	Marks
(a)	Understanding basic concepts	1
(b)	Understanding basic concepts	1

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(c)	Understanding basic concepts	1

|!|BCESC0020000010|!|

Fill in the blanks to complete the following paragraph. (6 marks)

Carbohydrates are organic compounds composed of carbon, hydrogen and (a)_____ atoms. They can be classified into monosaccharides, disaccharides and polysaccharides. Monosaccharides are all reducing sugars. They show positive result in the (b)_____test. Disaccharides are reducing sugars, except (c)_____. Polysaccharides include cellulose which is composed of a larger number of (d)_____ molecules combined by (e)_____. Cellulose is the component of (f)_____ in plant cells.

##

(a) oxygen (1)
(b) Benedict's (1)
(c) sucrose (1)
(d) glucose (1)
(e) condensation (1)
(f) cell wall (1)

(6 marks)

##

	Skills assessed	Marks
(a)	Understanding basic concepts	1
(b)	Understanding basic concepts	1
(c)	Understanding basic concepts	1
(d)	Understanding basic concepts	1
(e)	Understanding basic concepts	1
(f)	Understanding basic concepts	1

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|!|BCESB0020000011|!| ★ SI

The table below shows the composition of four types of sports drinks, A, B, C and D.

Sports drink	Ion content (mg / L)	Glucose content (mg / L)	Lipid content (mg / L)	Protein content (mg / L)
A	22	100	0	1.0
B	26	140	0	0.2
C	80	80	0.3	0
D	95	68	0.1	0

(a) Which energy drink has the greatest energy content? (1 mark)

(b) Describe how you would use a biochemical test to show that the drinks contain reducing sugars. (2 marks)

(c) Sports drinks are designed to be consumed during or after exercise. Ions are usually added in sports drinks.

(i) With reference to the body response during exercise, explain why ions are added in sports drinks. (3 marks)

(ii) Suggest ***one*** example of ion added in the sports drink. (1 mark)

##

(a) Sports drink B (1)

(b) Heat the drink samples with Benedict's solution. (1)
The formation of brick-red precipitates would show that the drinks contain reducing sugars. (1)

(c) (i) Our bodies sweat during exercise to increase heat loss. (1)
Ions are added to replace the ions lost in sweats during exercise. (1)
The replenishment of ions can help maintain water balance / osmotic balance in the body. (1)

(ii) sodium ion / chloride ion / potassium ion / other reasonable answer (1)

(7 marks)

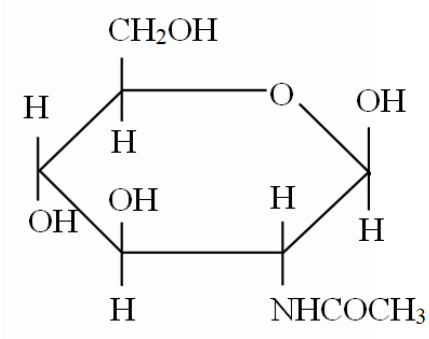
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##

	Skills assessed	Marks
(a)	Analysing data	1
(b)	Practical skills	2
(c)(i)	Applying concepts	3
(c)(ii)	Understanding basic concepts	1

|!|BCESA0020000012|!| ★★

Chitin is a type of polysaccharide found in the exoskeletons of insects and crustaceans. The following figure shows the basic unit of chitin.



(a) According to the figure, name the element that is present in chitin but ***not*** in other polysaccharides such as cellulose. (1 mark)

_____

(b) Name the chemical reaction in which the basic units are joined together to form chitin. (1 mark)

_____

(c) Chitosan is a polysaccharide derived from chitin. It can act as dietary fibre and is usually used as a dietary supplement for body slimming.

(i) Both chitosan and cellulose can act as dietary fibres. What is dietary fibre? (1 mark)

_____

(ii) According to the property of dietary fibres, describe how chitosan can be used to achieve body slimming. (3 marks)

_____

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##

(a) nitrogen (1)

(b) condensation (1)

(c) (i) Dietary fibre is the indigestible portion of food. (1)

(ii) Chitosan cannot be digested in our bodies, so it has no energy value to us. (1)

It fills the stomach and gives a sense of fullness to reduce food intake. /

It facilitates peristalsis to reduce nutrient absorption. (1)

The reduction of food intake / nutrient absorption may lead to less energy intake. (1)

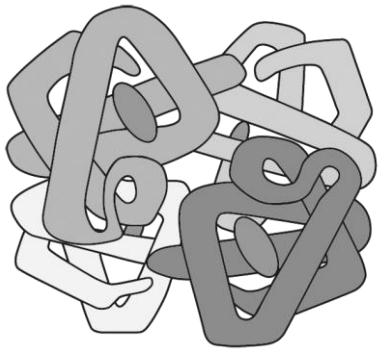
(6 marks)

##

	Skills assessed	Marks
(a)	Analysing data	1
(b)	Understanding basic concepts	1
(c)(i)	Understanding basic concepts	1
(c)(ii)	Applying concepts	3

|!|BCESA0020000013|!| ★★ **Cross topic**

The following diagram shows the structure of haemoglobin, a protein with a specific three-dimensional shape.



(a) Besides carbon, hydrogen and oxygen, name ***two*** other elements that are present in a haemoglobin molecule. (2 marks)

(b) Briefly describe the structure of a haemoglobin molecule. (2 marks)

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(c) What is the function of haemoglobin in our bodies? (1 mark)

(d) Haemoglobin is broken down in the liver to form amino acids and a pigment. Describe the fate of these amino acids. (2 marks)

##

(a) nitrogen (1)
iron (1)

(b) A haemoglobin molecule consists of four polypeptide chains folded together to form a compact structure. (1)
Each polypeptide chain encloses a haem group (an iron-containing component). (1)

(c) It carries oxygen in red blood cells. (1)

(d) The amino group of the amino acid is removed which is converted to urea. (1)
The remaining part of the amino acid is used to form carbohydrates or lipids. (1)

(7 marks)

##

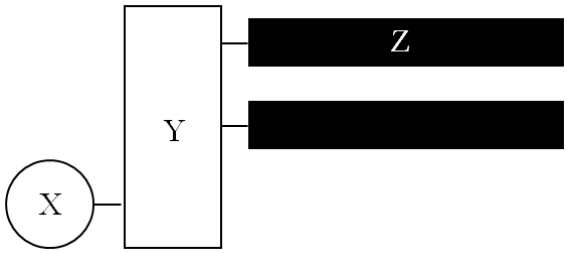
	Skills assessed	Marks
(a)	Understanding basic concepts	2
(b)	Understanding basic concepts	2
(c)	Understanding basic concepts	1
(d)	Understanding basic concepts	2

|!|BCESB0020000014|!| ★ **Cross topic**

(a) Coconuts are fruits that contain large amounts of lipid. Describe how you would use a test to show that a coconut contains lipid. (3 marks)

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(b) The diagram below shows the structure of a phospholipid molecule.



(i) What does the part of the molecule labelled Y represent? (1 mark)

(ii) Describe how the molecule shown in the diagram is different from a triglyceride molecule. (1 mark)

(iii) Phospholipids are arranged in a bilayer in the cell membrane. Explain how the properties of part Z contribute to this arrangement. (2 marks)

##

(a) Add a drop of coconut extract on a piece of filter paper. (1)
When the drop is dried, the lipids will leave a translucent spot, (1)
which will disappear after addition of organic solvent. (1)

(b) (i) glycerol (1)

(ii) Phospholipid has a phosphate group / (only) two fatty acids instead of three. (1)

(iii) Part Z (fatty acid) is hydrophobic / non-polar. (1)
They are attracted to fatty acid tails of other phospholipids. /
Fatty acid tails face away from water. (1)

(7 marks)

##

	Skills assessed	Marks
(a)	Practical skills	3
(b)(i)	Analysing data	1
(b)(ii)	Applying concepts	1
(b)(iii)	Applying concepts	2

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|!|BCESC0020000015|!|

Fill in the blanks to complete the following paragraph. (6 marks)

The basic units of nucleic acids are (a)_____. Each of them has a five-carbon sugar, a phosphate group and a (b)_____. In DNA, the five-carbon sugar is (c)_____. In RNA, the five-carbon sugar is (d)_____.

A DNA molecule consists of two chains. They coil around each other to form a (e)_____. (f)_____ formed between the bases can keep the DNA structure stable.

##

(a) nucleotides (1)
(b) nitrogenous base (1)
(c) deoxyribose (1)
(d) ribose (1)
(e) double helix (1)
(f) Hydrogen bonds (1)

(6 marks)

##

	Skills assessed	Marks
(a)	Understanding basic concepts	1
(b)	Understanding basic concepts	1
(c)	Understanding basic concepts	1
(d)	Understanding basic concepts	1
(e)	Understanding basic concepts	1
(f)	Understanding basic concepts	1

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